pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Metadata Filters and Many Documents

WEEK 2 · DAY 4

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: a corpus of four

Four documents, one corpus

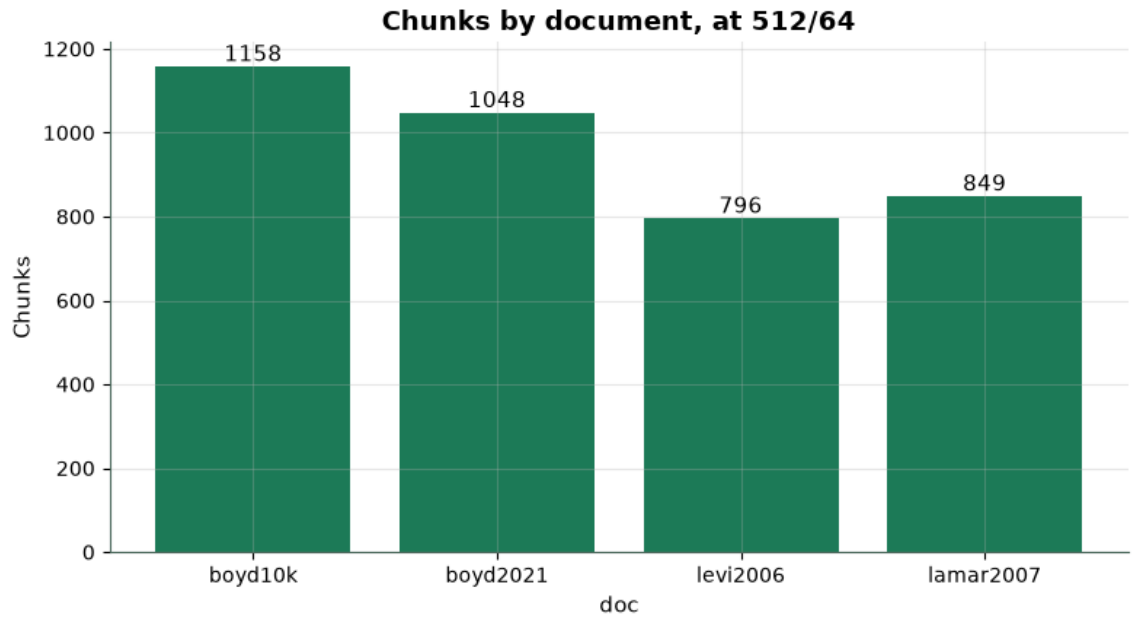
Corpus	3,851 chunks at 512/64
Documents	a 10-K and three indentures
Questions	25 development questions
Answers	8 saved answers, replayed
Held-out	not opened until day 11

Today

1. One corpus, four documents
2. Metadata as columns, filters as masks
3. A question with no filter
4. Filter before search
5. The join the corpus allows
6. A question the corpus cannot answer
7. The wrong document

Say which document a question is about with a filter, and read the document in the citation.

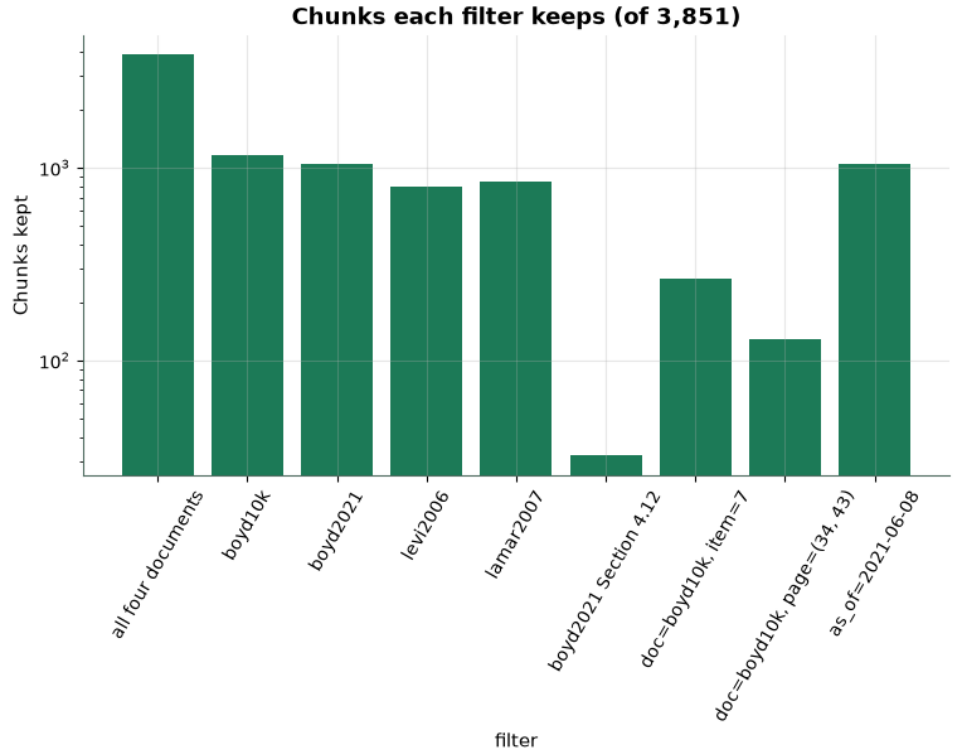
One corpus, four documents



document	cut inside	cuts	chunks	as of
boyd10k	items	24	1158	2025-12-31
boyd2021	sections	108	1048	2021-06-08
levi2006	sections	86	796	2006-03-17
lamar2007	sections	123	849	2007-10-11

A 10-K is cut inside its items, an indenture inside its sections. The last section of each indenture runs on into its exhibits.

Metadata as columns, a filter as a mask



filter_chunks(chunks, **metadata)

```
doc

item  (section)

page  (one, or first and last)

as_of  (a value or a list)
```

Returns one True or False per chunk. It searches nothing. The search is handed the rows the mask keeps. An unknown field raises ValueError.

The scopes differ by a factor of 100: 3,851 chunks for all four documents, 32 for one section.

What is the interest rate on the senior notes?

The answer in the saved run

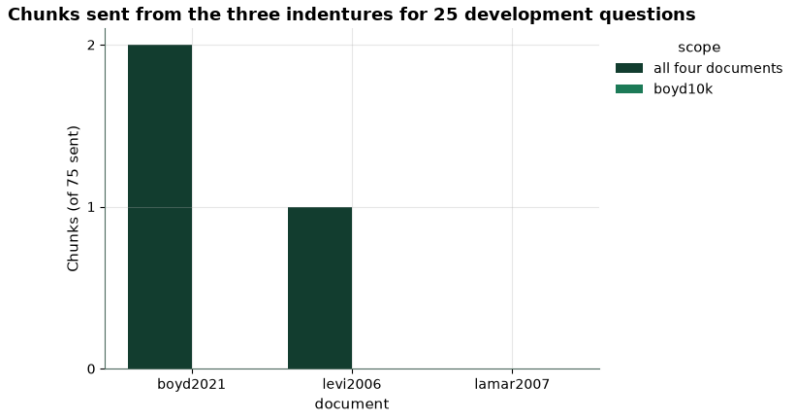
The interest rate on the 6 5/8% Senior Subordinated Notes Due 2015-Series C is 6 5/8% per annum [lamar2007; Section 12.16; c0769] and [lamar2007; Section 12.16; c0700]. The interest rate on the 4.750% Senior Notes due June 2031 is 4.750% per annum [boyd10k; Item 8; p. 77; c0844].

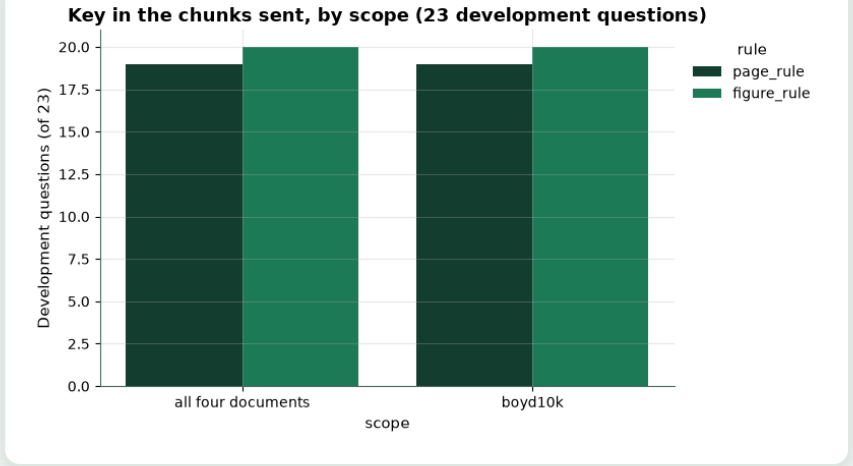
The three chunks behind it

citation	score
[lamar2007; Section 12.16; c0769]	7.21
[lamar2007; Section 12.16; c0700]	7.19
[boyd10k; Item 8; p. 77; c0844]	6.43

Documents among the chunks: lamar2007, boyd10k; cited: lamar2007, boyd10k. The question named no issuer.

Same question, two filters





With a filter

10-K only: cites boyd10k, 4.750%

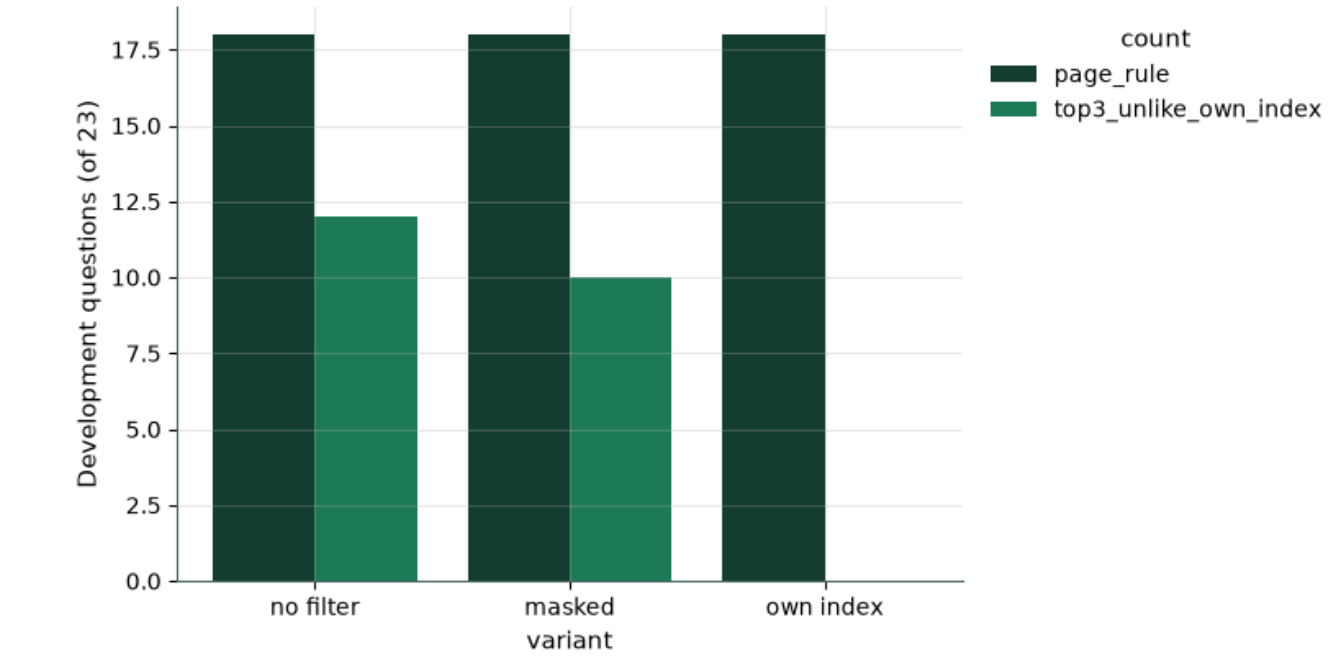
Boyd indenture only: cites boyd2021, 4.750%

```
page rule / figure rule: all four documents 19/20; boyd10k 19/20

chunks from other documents: all four documents 3; boyd10k 0; in dev08, dev09, dev10
```

Identical chunks with and without the filter: 22 of 25. A filter changes the result for the questions the unwanted documents can answer.

The index follows the rows



Three ways to score a filtered search

variant	page	other	unlike
no filter	18	2	12
masked	18	0	10
own index	18	0	0

Key page found in the top 3: no filter 18; masked 18; own index 18. Top 3 differ between masked and own index for 10 of 23 questions.

Build the index on the rows the search reads. It costs a fifth of a second.

Two documents, two filtered questions

Each answer cited to its own document

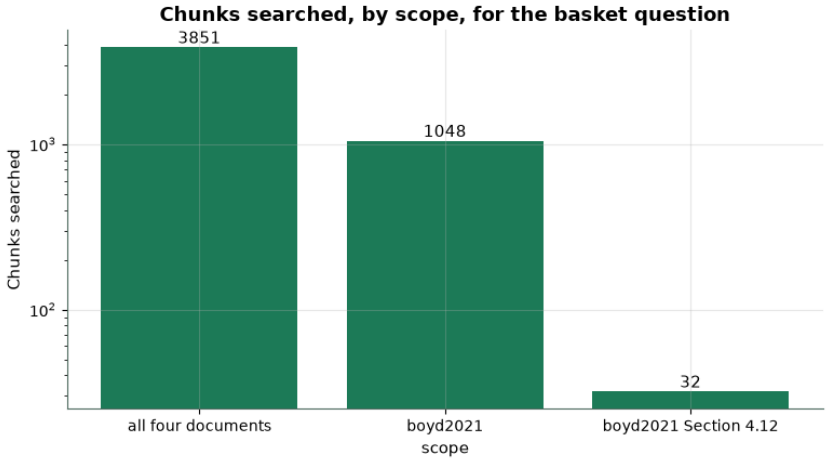
10-K: The 4.750% Senior Notes due 2031 were issued on June 8, 2021, in the aggregate principal amount of $900.0 million [boyd10k; Item 7; p. 39; c0441], [boyd10k; ...

Boyd indenture, Section 4.12: not to exceed the greater of (A) $430,000,000 and (B) 7.5% of the Company's Consolidated Total Assets, calculated on a Pro Forma Basis [boyd2021; Section 4.12; c0549].

```
boyd10k                 key $900.0 million      match

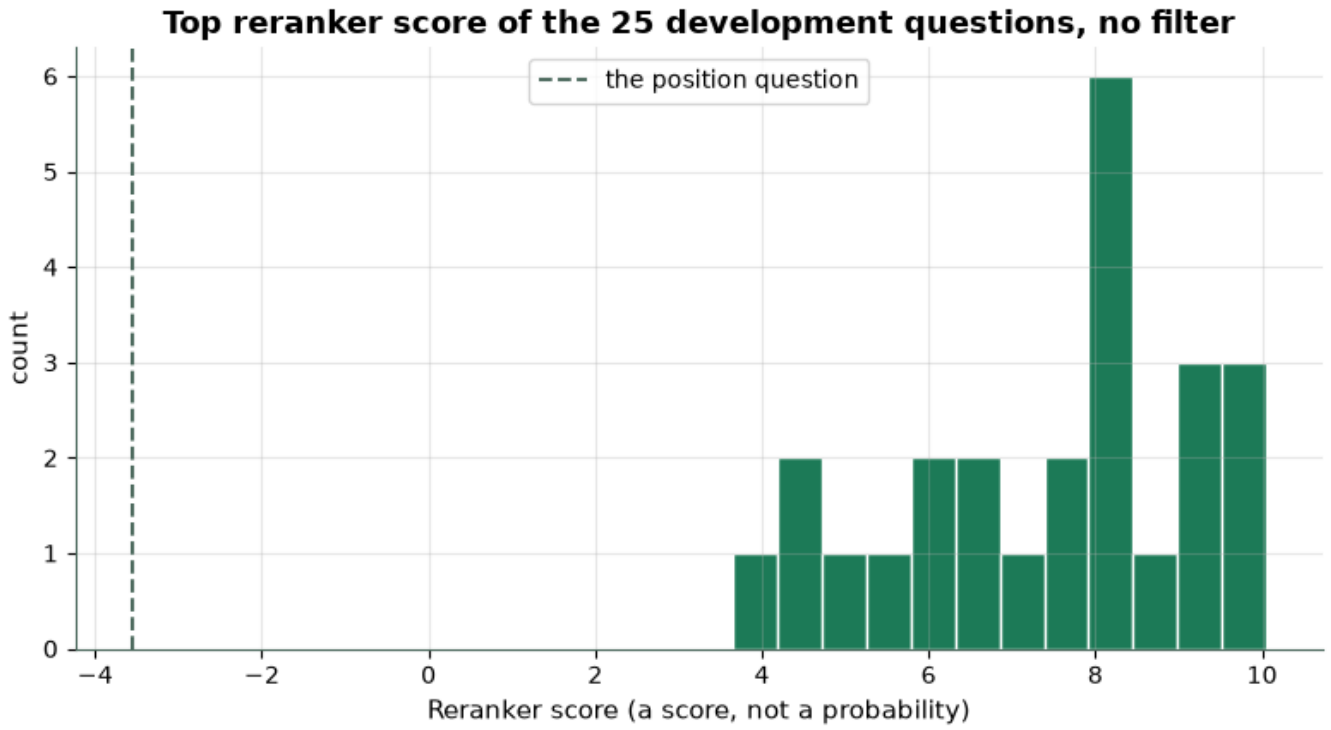
boyd2021 Section 4.12   key $430,000,000; 7.5%  match
```

Chunk with the figures among the three sent: all four documents: not in the three; boyd2021: not in the three; boyd2021 Section 4.12: 3



Figures two public documents state, set side by side. Nothing is said about which is larger.

A position the corpus does not hold



Reranker scores

The position question's three chunks: -3.56, -3.77, -3.79

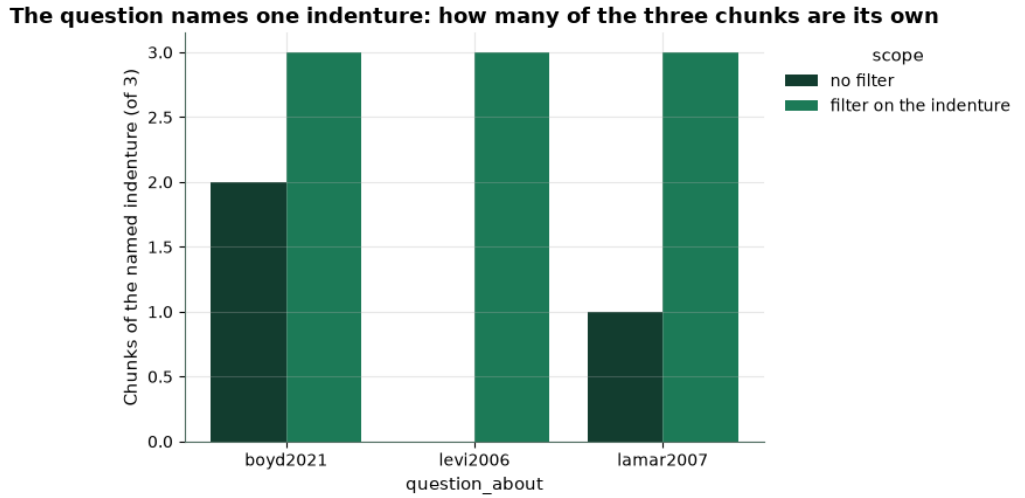
Answerable questions' top score: 3.66 to 10.05.

Questions the 10-K cannot answer: dev24 7.21; dev25 8.10

Answer: NOT IN CORPUS

A score says how well a chunk matches, not whether the question can be answered from it. It is not a probability.

A correct figure from the wrong document



The Levi Strauss question

No filter: "not less than 30% in aggregate principal amount of the Notes then Outstanding" [boyd2021; Section 6.02; c0660].

Filter on levi2006: "at least 25% in aggregate principal amount of the Securities then outstanding" [levi2006; Section 10.13; c0770].

```
no filter           cites boyd2021   asked-about document cited: no  grade: no match
filter on levi2006  cites levi2006   asked-about document cited: yes grade: match
```

Chunks of the named indenture among the three: no filter boyd2021 2; levi2006 0; lamar2007 1; filter boyd2021 3; levi2006 3; lamar2007 3. The citation names the document; a key may not.

Excel side by side, and Git

Excel side by side (tested in Excel)

`=COUNTIFS(chunks[B],"boyd2021")`	chunks of boyd2021: 1,048, equal to Python for all four
`=COUNTIFS(chunks[B],"boyd2021",chunks[C],"4.12")`	a document and a section: 32
`=ROWS(FILTER(chunks[A],chunks[B]="lamar2007"))`	rows kept for lamar2007: 849
`=TAKE(SORTBY(FILTER(...),FILTER(...),-1),3)`	filter first: 3 ids, the same as Python
`=FILTER(TAKE(SORTBY(...),3),TAKE(...)="lamar2007")`	top 3 first, then filter: 1 row

A section such as 4.10 pasted into a General cell becomes the number 4.1: MATCH("4.10",...) returns not found. Paste as Text.

Git: corpus_manifest.json, then the 10-K's rows: changed documents, chunks, matrix_sha256; 47 tests.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the function filters after the search, so it returns fewer than k chunks.

Code that runs is not the same as code that is right.

What you can do now

- A corpus: each document cut inside its own units
- A filter is a row mask, applied before the search
- Build the index on the rows the search reads
- Read the document in the citation, not only the figure
- A score is not a refusal rule; NOT IN CORPUS is the model's job

Week 2, Day 5 Case study: prompts that must cite